



আন্তর্জাতিক ইসলামী বিশ্ববিদ্যালয় চট্টগ্রাম
الجامعة الإسلامية العالمية سبّاغونغ
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Lab Final Project Manual

Machine Learning and Data Mining Lab (CSE-4878)

Department of Computer Science and Engineering

Credit Hours: 1 | Contact Hours: 2 | Semester: 7th

1. Project Overview

1.1 Objective

This final project requires students to conduct a **complete project research study** using **text data**. Students will progress from data collection to a standard IEEE conference paper. The goal is to apply machine learning or data mining techniques to solve a real-world text-based problem, such as:

- Sentiment analysis
- Fake news detection
- Spam classification
- Topic modeling
- Hate speech detection
- Suicide risk detection from social media, and so on.

1.2 Research Expectation

The final output must be of sufficient quality to be submitted to an IEEE conference (e.g., equivalent to any student-friendly IEEE conference).

2. Alignment with Course Learning Outcomes (CLOs)

CLO	Description	How This Project Addresses the CLO
CLO1	Demonstrate statistical data analysis with modern tools	Exploratory Data Analysis (EDA) on text: word frequency distributions, class balance, length statistics
CLO2	Apply data preprocessing to improve model performance	Text cleaning, tokenization, stopword removal, stemming/lemmatization, vectorization (TF-IDF, embeddings)

CLO3	Design and evaluate ML models using appropriate algorithms and metrics	Build classifiers (SVM, Random Forest, Naive Bayes) or clustering models; report accuracy, F1-score, precision, recall, confusion matrix
CLO4	Analyze fairness, bias, and legal aspects in algorithmic decision-making	Evaluate model performance across text subgroups; discuss data bias, ethical considerations, and fairness limitations
CLO5	Assess societal/environmental impact and document through reports/presentations	Write structured IEEE paper; discuss real-world impact and environmental cost (e.g., CO ₂ estimation)

3. Deliverables

#	Deliverable	Format	Submission Mode
1	Python code with comments	Jupyter Notebook (.ipynb) / Colab	GitHub + Preferred LMS
2	IEEE-style research paper	PDF (4–6 pages)	Preferred LMS
3	Presentation (live)	PPT/PDF (5 min)	Preferred LMS
4	GitHub repository (private with instructor access)	Link	Preferred LMS

4. Dataset Guidelines (Text Data Only)

Self-Collected Data (Higher Impact)

- Scrape social media or collect secondary data
- Collect customer reviews from e-commerce sites
- Use Mendeley (Highly recommended), Kaggle text datasets (e.g., Disaster Tweets, AG News)

Minimum Requirement:

- Classification: 1,000 labeled text samples

5. Step-by-Step Research Pipeline

Phase 1: Problem Definition & Ethics (Week 1)

Task: Write a clear problem statement.

Example: *"Detect depressive language in Reddit posts to enable early mental health intervention."*

Ethical Check:

- Will your model harm any group?
- Is the data biased or non-consensual?
- Did you anonymize personally identifiable information (PII)?

Deliverable: 1-page proposal (PDF) with:

- Problem statement
 - Dataset source
 - Potential societal impact
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Phase 2: Data Collection & Exploratory Analysis (Week 2)

Tasks (CLO1):

- Load text data into Pandas DataFrame
- Perform univariate analysis:
 - Distribution of text lengths
 - Most frequent words (word cloud)
 - Class balance (for classification)
- Visualize using Matplotlib and Seaborn

Deliverable: Code + visualizations in notebook.

Phase 3: Text Preprocessing (Week 3)

Mandatory steps (CLO2):

1. Convert to lowercase
2. Remove punctuation, numbers, special characters
3. Tokenization (split into words)
4. Remove stopwords
5. Apply stemming (Porter) or lemmatization (WordNet)
6. Remove rare words (optional)

Numerical Conversion (choose one):

- **Easy:** TF-IDF Vectorizer (Scikit-learn)
- **Publishable:** Word2Vec, GloVe, or BERT embeddings

Deliverable: Cleaned and vectorized dataset ready for modeling.

Phase 4: Model Building & Evaluation (Weeks 4–5)

Implement at least two algorithms (CLO3):

Task Type	Algorithms to Try (Using recent algorithms is preferable)
Classification	Logistic Regression, SVM, Random Forest, Naive Bayes
Clustering	K-Means, DBSCAN (on text vectors)
Topic Modeling	LDA (Latent Dirichlet Allocation)

Evaluation Metrics (use all that apply):

- Accuracy, Precision, Recall, F1-score
- Confusion Matrix
- ROC Curve (for binary classification)
- For clustering: Silhouette Score, Davies–Bouldin Index (Optional)

Baseline Comparison:

Compare your best model against a simple baseline.

Deliverable: Trained models + evaluation tables/figures.

Phase 5: Bias & Fairness Analysis (Week 6)

Required for CLO4

Simple methods:

- Compare model performance across different text subgroups (e.g., short vs. long texts, by source, by topic)
- Train on balanced vs. imbalanced data and compare results

Questions to answer in your paper:

- Does your model perform worse for certain text types?

- Could your model amplify stereotypes?
- What legal or ethical issues arise from deployment?

Optional tools: Aequitas, Fairlearn, or manual group-split analysis.

Deliverable: Written analysis (minimum 1 paragraph in Discussion section).

Phase 6: Societal & Environmental Impact (Week 6)

Required for CLO5

Societal Impact:

- Who benefits from this model?
- Who could be harmed?
- Can the model be misused?

Deliverable: Written impact statement in Conclusion/Discussion.

Phase 7: Write IEEE formatted Paper (Final Report) (Week 7)

Use the IEEE template provided in Section 6.

Deliverable: Draft PDF (4–6 pages).

Phase 8: Final Submission & Presentation (Week 8)

Deliverables:

- Final cleaned Jupyter Notebook/Colab
 - Final IEEE paper (PDF)
 - Presentation slides (5 minutes)
 - GitHub link
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6. IEEE Publication Guidelines

6.1 Paper Format

Use the **IEEE Conference Template** (double column, 10pt font).

Download from: <https://www.ieee.org/conferences/publishing/templates.html>

<https://www.overleaf.com/gallery/tagged/ieee-official>

6.2 Required Sections and Page Limits

Section	Approx. Pages	Required Content
Abstract	0.5	Problem, method, key result, impact (150–200 words)
Introduction	0.75	Problem, motivation, 3–4 bullet-point contributions
Related Work	0.5	Cite 10–15 papers from Google Scholar (Pre-prints are not acceptable)
Methodology	1.5	Data description, preprocessing, model architecture, hyperparameters
Experiments & Results	1.5	Dataset statistics, evaluation metrics, tables, figures (confusion matrix, ROC)
Discussion	0.75	Bias, fairness, limitations, future work
Conclusion	0.25	Summary + real-world impact statement
References	0.25	10–15 IEEE-style references

6.3 Mandatory Figures and Tables

- Table 1: Dataset statistics (classes, samples per class, avg. text length)
- Table 2: Model comparison (accuracy, precision, recall, F1-score)
- Figure 1: Word cloud or top feature visualization
- Figure 2: Confusion matrix
- Figure 3: ROC curve (for binary classification – optional but encouraged)

6.4 Example Title (IEEE Style)

"Detecting Depression Indicators in Social Media Text: A Comparative Study of SVM and Random Forest"

6.5 Example Reference (IEEE Format) – Auto Formatting in Overleaf

[1] J. Devlin, M. Chang, K. Lee, and K. Toutanova, "BERT: Pre-training of deep bidirectional transformers for language understanding," in Proc. NAACL, 2019, pp. 4171–4186.

7. Grading Rubric (SEE – 50 Marks)

Component	Marks	CLO Mapped
Problem definition & ethical consideration	5	CLO4

Data preprocessing & exploratory data analysis	10	CLO1, CLO2
Model implementation (at least two algorithms)	15	CLO3
IEEE paper (structure, clarity, references, figures)	10	CLO5
Presentation (5 minutes) + GitHub repository	10	CLO5
Total	50	

8. Recommended Tools & Libraries

Task	Recommended Library	Ease of Use
Data handling	Pandas, NumPy	Easy
Text preprocessing	NLTK, spaCy, re	Easy
Vectorization	Scikit-learn (TfidfVectorizer, CountVectorizer)	Easy
Machine learning models	Scikit-learn (all classifiers)	Easy
Deep learning (optional)	TensorFlow / Keras (simple FFNN)	Medium
Visualizations	Matplotlib, Seaborn, WordCloud	Easy
Paper writing	Overleaf (LaTeX) + IEEE template	Easy

9. Submission Checklist

Before final submission, verify:

- Jupyter Notebook runs end-to-end without errors
- All code cells are commented
- Dataset is included via link (Google Drive/Mendeley/Kaggle) – do not upload large files
- GitHub repository contains README with setup instructions
- IEEE paper has all sections: Abstract to References
- Societal and environmental impact statement is present
- At least two ML models are implemented and compared
- Confusion matrix and evaluation table are included
- Presentation (5 min) is accessible